Cheatsheet AZ-305 — 03 · Azure Storage complete

1. Redundancy: copies and what it protects against

Type	Copies	Where	Protects against	Read from secondary
LRS	3	1 datacenter	Disk / node failure	—
ZRS	3	3 zones (same region)	Failure of **one datacenter / zone**	—
GRS	6	3 LRS primary + 3 LRS paired region (async)	**Region** failure	✗
RA-GRS	6	same as GRS	Region failure	✓ -secondary endpoint
GZRS	6	3 ZRS primary + 3 LRS secondary	Zone **and** region	✗
RA-GZRS	6	same as GZRS	Zone and region	✓

- "datacenter failure" → **ZRS** is enough. "region failure" / "maximize resiliency" → **GRS / GZRS**.
- "replicates" ≠ "read from replica": if reading from the secondary is not required, **GRS** (cheaper than RA-GRS).
- **Premium (SSD) only supports LRS and ZRS.** Never GRS.

2. Account failover cycle (customer-managed)

```
Before: configure GRS/RA-GRS (no secondary, no failover)
   → Initiate failover (the secondary becomes primary; the endpoint does not change for apps)
   → The account is automatically left in LRS
   → Reconfigure GRS to be able to fail back
   → Failback once re-replication completes
```

Exam sequence: **4 (GRS before) → 2 (failover) → 1 (GRS after)**. ZRS never appears.

3. Account types

Type	Performance	Services	Hot/Cool/Archive tiers	Redundancy	When
General-purpose v2 (StorageV2)	Standard (HDD)	Blob, Files, Queue, Table, ADLS Gen2	✓	All	**Default** for everything new
General-purpose v1	Standard	Blob, Files, Queue, Table	✗	Limited	Legacy; distractor
Blob Storage (legacy)	Standard	Blobs only	✓	All	Legacy; StorageV2 replaces it
BlockBlobStorage (Premium block blobs)	**Premium SSD**	Block / append blobs	✗ (single tier)	**LRS / ZRS**	Maximum transactions and minimum latency on blobs
Premium page blobs	Premium	Page blobs	✗	LRS / ZRS	Unmanaged VM disks
Premium file shares (FileStorage)	Premium	Azure Files (SMB/NFS)	✗	LRS / ZRS	High-performance file shares

- **GPv1 + Premium** does not exist (invalid option).
- **Data Lake Storage Gen2** = StorageV2 (or Premium block blob) with **hierarchical namespace**: ABFS/Hadoop driver, **POSIX ACLs** per folder and file, real directories.
- "Hadoop", "HDFS", "POSIX ACL" → GPv2 with hierarchical namespace.

4. Blob access tiers

Tier	Storage cost	Access cost	Read latency	Minimum retention	Use
Hot	High	Low	ms	—	Frequent access, active uploads/downloads
Cool	Medium	Medium	ms	30 days	Infrequent access, immediate access
Cold	Low	High	ms	90 days	Very infrequent, immediate access
Archive	Minimum	Maximum	**Hours (rehydration, up to 15 h)**	180 days	Almost never; can wait

- "available within 24 hours" / "rarely accessed" + "minimize cost" → **Archive** (blobs only, GPv2 / Blob Storage; not Files, not GPv1).
- "optimized for uploads/downloads" + "lowest cost" → **Hot** on Standard (the saving comes from Standard vs Premium, not from Cool).
- The account's **default tier** does not matter if each blob is explicitly set to Archive.
- **Lifecycle management** moves blobs between tiers and deletes them by age rules.

5. Data protection on blobs

Feature	What it does	Time direction	Prerequisites
Soft delete (blob / container)	Keeps deleted items for N days	Recover what was deleted	—
Versioning	Saves every version on overwrite	Recover previous versions	—
Change feed	Ordered, immutable log of all changes	Know what changed and when	—

Feature	What it does	Time direction	Prerequisites
Point-in-time restore	Return a container to a point in time	Go back	**Versioning + soft delete + change feed**
Immutable storage (WORM)	Prevents modifying / deleting during retention	Prevent changes going forward	Versioning (at version level)
Object replication	Async copy of blobs to another account	Replicate going forward	Versioning + change feed
Resource lock (CanNotDelete)	Prevents deleting the resource (account), not the blobs	ARM resource protection	—

- **Immutable: time-based retention** (X years; **Locked** for regulatory, Unlocked can be shortened) and **legal hold** (until removed). "new data can be written, modification prevented for 3 years" → time-based retention.
- PITR and immutability are incompatible on the same container.

6. Access and security

Mechanism	What it is	When
Account keys	Two master keys	Avoid; rotate; never in clients
SAS (account / service / user delegation)	Token with permissions, expiry, IP	Temporary access for clients; **stored access policy** = centralized revocation
Entra ID RBAC (Storage Blob Data Reader/Contributor)	Roles on the data plane	Preferred for apps with managed identity
Private Endpoint	Private IP in the VNet + Private DNS zone	"no public endpoint", access from on-prem via ExpressRoute/VPN
Service endpoint	The subnet reaches the service over the Microsoft network; the service keeps a public IP	When a private IP is not required
Account firewall	Allowed IPs / VNets	Complement
SSE	Encryption at rest always on; **PMK** by default or **CMK** in Key Vault	"manage own keys" → CMK
Secure transfer required	Forces HTTPS	Encryption in transit

7. Azure Files and File Sync

- **Azure Files:** SMB / NFS; tiers Premium, Transaction Optimized, Hot, Cool (**no Archive**). Identity: Entra Domain Services, on-prem AD DS or Entra Kerberos.
- **Azure File Sync:** local cache on Windows Server + cloud tiering; continued file access if a site goes down.
- **Azure File Sync ≠ backup**: to copy, use Azure Backup for file shares.

8. Bulk transfer

Tool	Mechanism	When
AzCopy	Copy over the network	Moderate volumes, scripting
Import/Export	You ship **your own disks** encrypted with BitLocker	TB with little bandwidth, own hardware
Data Box (Disk / Data Box / Heavy)	**Microsoft ships a device**	Tens of TB to PB
Data Box Gateway	Virtual appliance that sends to Azure continuously	Continuous transfer from on-prem
Data Share	Share snapshots with **another organization** in a governed way	External partner, another tenant

9. Decision tree (account + redundancy + tier)

```
Maximum performance / minimum latency on blobs?
├─ Yes → BlockBlobStorage Premium → ZRS (datacenter) or LRS redundancy
└─ No → StorageV2 Standard
        ├─ Maximum resiliency → GRS / RA-GRS (RA only if reading is required)
        ├─ Datacenter only → ZRS
        ├─ Hadoop / POSIX → hierarchical namespace (ADLS Gen2)
        └─ Tier: frequent → Hot · infrequent immediate → Cool · can wait hours → Archive
```

10. Signals → answer

Wording	Answer
"highest transaction rate, lowest latency"	BlockBlobStorage Premium + ZRS
"lowest cost per GB" + "uploads/downloads"	GPv2 Standard Hot
"maximize data resiliency"	StorageV2 + RA-GRS (never Premium)
"available within 24 hours"	Archive
"prevent modification for N years"	Immutable, time-based retention (Locked)
"point-in-time restore" (versioning and soft delete already enabled)	Change feed
"fail over and fail back"	GRS before → failover → GRS after

Wording	Answer
"Hadoop-compatible", "POSIX ACLs"	GPv2 + hierarchical namespace
"REST" + "20 independent tables" + "second region" + "min cost"	Table Storage + GRS
"public endpoint must be prevented"	Private Endpoint
"manage own encryption keys"	SSE with CMK (Key Vault)
"ship physical disks", "own hardware"	Import/Export
"share snapshots with partner company"	Data Share

11. Mistakes to avoid

1. Premium + GRS → does not exist.
2. RA-GRS when nobody asks to read from the secondary → GRS is cheaper.
3. Archive on Azure Files or on GPv1 → does not exist.
4. Cool for "optimized for uploads/downloads" → penalizes transactions; Hot.
5. Forgetting that failover leaves the account in LRS.
6. Confusing Data Box (Microsoft's device) with Import/Export (your own disks).